

1. A 47-year-old woman comes to the physician because of a 1-week history of progressive pain and swelling of the left knee. She has pain with walking and at rest. There is no history of trauma. She has hypertension, type 1 diabetes mellitus, and asthma. Her current medications are insulin, a β -adrenergic blocking agent, and a daily inhaled corticosteroid. She is 158 cm (5 ft 2 in) tall and weighs 136 kg (300 lb); BMI is 55 kg/m². Her temperature is 37.8°C (100.1°F); other vital signs are within normal limits. The knee is tender, swollen, warm, and erythematous. There is severe pain with passive and active range of motion. Plain x-rays of the knee show periarticular osteopenia; the joint space is narrowed medially. Which of the following is the most appropriate next step in management?

- ☐ A) Measurement of serum uric acid concentration
- ☐ B) MRI of the knee
- ☐ C) Aspiration of the knee joint
- ☐ D) Intra-articular injection of corticosteroids
- ☐ E) Naproxen therapy



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2. A 16-year-old girl comes to the physician for a routine health maintenance examination. She says that she began following a vegan diet (no dairy or animal products) 2 months ago. She takes no medications or dietary supplements. She is 165 cm (5 ft 5 in) tall and weighs 50 kg (110 lb); BMI is 18 kg/m². Vital signs are within normal limits. Examination shows no abnormalities. Supplementation with which of the following is the most appropriate recommendation for this patient?

- ☐ A) Calcium
- ☐ B) Folic acid
- ☐ C) Magnesium
- ☐ D) Manganese
- ☐ E) Vitamin A



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3. A 62-year-old man returns to the physician because of a 4-month history of progressive difficulty swallowing. This symptom initially occurred only when swallowing large pills or solid food, but now he also has had difficulty swallowing liquids for more than 2 months. He says that he was evaluated by the physician and underwent a barium swallow 2 months ago but has not received the results. Review of the office chart shows no test results or notation to schedule a follow-up visit. The radiology facility is contacted and reports that the results were mailed to the office immediately after the study 2 months ago and show evidence of esophageal carcinoma. Which of the following is most likely to have prevented the delay in this patient's care?
- ☐ A) Asking the radiology facility to fax results directly to the office instead of mailing them
 - ☐ B) Encouraging the patient to return for a follow-up visit to discuss the test results
 - ☐ C) Keeping a log of ordered tests that require follow-up by the office
 - ☐ D) Referring all patients with gastrointestinal problems to specialists
 - ☐ E) Retraining office staff on proper filing and notification procedures



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4. A 67-year-old woman comes to the physician because of constant, severe, nonradiating low back pain that began 3 days ago after she changed her bedsheets. She has chronic asthma treated with an albuterol inhaler, prednisone, and theophylline. Menopause occurred 15 years ago; she has never received hormone replacement therapy. Her temperature is 36.6°C (97.8°F), pulse is 90/min, respirations are 20/min, and blood pressure is 148/90 mm Hg. There is decreased range of motion of the lumbar spine and point tenderness over L2. Neurologic examination shows no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Acute lumbosacral strain
- ☐ B) Ankylosing spondylitis
- ☐ C) Compression fracture
- ☐ D) Herniated nucleus pulposus
- ☐ E) Spondylolisthesis



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5. A 45-year-old man with type 2 diabetes mellitus and hypertension comes to the physician as a new patient. He has no other history of serious illness. He takes metformin and hydrochlorothiazide. He has smoked one pack of cigarettes daily for 20 years. His blood pressure is 142/82 mm Hg. The remainder of the examination shows no abnormalities. Laboratory studies from 1 month ago show:

Hemoglobin A _{1c}	7.8%
Serum	
Na ⁺	142 mEq/L
K ⁺	4.3 mEq/L
Cl ⁻	109 mEq/L
Urea nitrogen	14 mg/dL
Glucose	135 mg/dL
Creatinine	0.7 mg/dL
Cholesterol, total	214 mg/dL
HDL-cholesterol	50 mg/dL
LDL-cholesterol	120 mg/dL
Triglycerides	142 mg/dL

Which of the following modifications would be most beneficial to this patient?

- ☐ A) Smoking cessation
- ☐ B) Adding lisinopril therapy to the medication regimen
- ☐ C) Adding pravastatin therapy to the medication regimen
- ☐ D) Increasing the dosage of hydrochlorothiazide
- ☐ E) Increasing the dosage of metformin



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6. A 6-year-old boy is brought to the physician because of a 2-week history of scaling of his scalp. The family has no pets; the child attends an after-school day-care center, which has a pet turtle. The other children at the day-care center are asymptomatic. A photograph of the scalp is shown. Which of the following is the most appropriate pharmacotherapy to prevent spread to the children in the day-care center?

- ☐ A) Oral griseofulvin for the patient only
- ☐ B) Oral griseofulvin for the patient and his classmates
- ☐ C) Topical hydrocortisone for the patient only
- ☐ D) Topical hydrocortisone for the patient and his classmates
- ☐ E) Topical selenium sulfide for the patient only
- ☐ F) Topical selenium sulfide for the patient and his classmates



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7. A 72-year-old man comes to the physician as a new patient. He recently moved to the area. He feels well. He has hypertension and hypercholesterolemia. Current medications include ramipril, atorvastatin, and aspirin. Vital signs are within normal limits. Examination shows a harsh right carotid bruit. Neurologic examination shows no focal findings. Carotid Doppler ultrasonography shows 30% stenosis on the right and 10% stenosis on the left. Which of the following is the most appropriate next step in management?
- ☐ A) Continue his current regimen
 - ☐ B) Add clopidogrel to the regimen
 - ☐ C) Carotid angiography
 - ☐ D) Magnetic resonance angiography
 - ☐ E) Carotid endarterectomy



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8. A 57-year-old man comes to the physician because of a 6-month history of cough productive of gray to light yellow sputum, shortness of breath on exertion, and occasional wheezing. He has not had chest pain. He has smoked one and a half packs of cigarettes daily for 38 years. His pulse is 78/min, respirations are 20/min, and blood pressure is 140/88 mm Hg. Breath sounds are decreased; occasional wheezing and fine scattered crackles are heard. Cardiac examination shows normal findings. There is trace pedal edema with no jugular venous distention or hepatojugular reflux. Laboratory findings are within the reference range except for a hemoglobin concentration of 17.6 g/dL. An x-ray of the chest shows flattened diaphragms. An ECG shows right axis deviation. Which of the following is the most appropriate next step in management?
- ☐ A) Recommend participation in a smoking cessation program
 - ☐ B) Inhaled cromolyn therapy
 - ☐ C) Inhaled fluticasone therapy
 - ☐ D) Oral amoxicillin therapy



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9. A 25-year-old woman who is a schoolteacher has a positive PPD skin test at a routine examination. She was exposed to a person with active tuberculosis when she was a child; however, all previous PPD skin tests since that time have been negative. Her last PPD skin test was 5 years ago. An x-ray of the chest shows no abnormalities. Which of the following is the most appropriate next step in management?

- ☐ A) Repeat PPD skin test in 10 weeks
- ☐ B) Repeat PPD skin test in 1 year
- ☐ C) Second strength PPD skin test (250 tuberculin units) now
- ☐ D) Treatment of latent tuberculosis now
- ☐ E) Treatment of active tuberculosis now
- ☐ F) No testing or therapy indicated



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10. A 22-year-old primigravid woman at 34 weeks' gestation comes to the physician because of a 1-week history of burning pain in her right thumb, index, and middle fingers and aching in her right forearm. Pregnancy has been uncomplicated. Her only medication is a prenatal vitamin. Muscle strength and sensation are intact in the right arm and hand. Which of the following is the most likely cause of this patient's symptoms?

- ☐ A) Cervical radiculopathy
- ☐ B) Median nerve entrapment
- ☐ C) Meralgia paresthetica
- ☐ D) Multiple sclerosis
- ☐ E) Radial nerve palsy
- ☐ F) Vitamin B₆ deficiency



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11. A 16-year-old boy comes for an examination prior to participation in school sports. A painless mass is palpated in the scrotum; the mass decreases while supine and has a bag-of-worms sensation. The boy states that he did not notice it previously. Which of the following is the most likely diagnosis?

- ☐ A) Epididymitis
- ☐ B) Hydrocele
- ☐ C) Inguinal hernia
- ☐ D) Leydig cell tumor
- ☐ E) Teratoma of the testis
- ☐ F) Torsion of the testis
- ☐ G) Varicocele



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12. A 42-year-old man comes to the physician because of a 2-month history of severe leg pain, especially at night. He has poorly controlled type 1 diabetes mellitus complicated by diabetic nephropathy. One month ago, his urine creatinine clearance was 20 mL/min. Current medications include insulin and captopril. Sensation to pain and light touch is decreased over the lower extremities in a stocking-glove distribution. His hemoglobin A_{1c} is 10.2%. In addition to tighter glucose control, which of the following is the most appropriate next step in management of this patient's leg pain?

- ☐ A) Topical capsaicin therapy
- ☐ B) Oral acetaminophen with codeine therapy
- ☐ C) Oral amitriptyline therapy
- ☐ D) Oral ibuprofen therapy
- ☐ E) Oral morphine therapy



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13. An asymptomatic 27-year-old woman comes for a routine health maintenance examination. She works as a lawyer. Her temperature is 37.2°C (98.9°F), pulse is 96/min, respirations are 16/min, and blood pressure is 144/82 mm Hg. Serum studies show a thyroxine (T_4) concentration of 14 µg/dL, a triiodothyronine (T_3) concentration of 250 ng/dL, and a thyroid-stimulating hormone concentration of 0.1 µU/mL. A radioactive thyroid scan shows increased uptake and a diffusely enlarged gland. Which of the following is the most likely cause of this patient's findings?

- ☐ A) Acute suppurative thyroiditis
- ☐ B) Chronic lymphocytic thyroiditis (Hashimoto disease)
- ☐ C) Drug-induced hypothyroidism
- ☐ D) Euthyroid sick syndrome
- ☐ E) Exogenous administration of thyroid hormone
- ☐ F) Hyperthyroidism
- ☐ G) Multiple endocrine neoplasia syndrome
- ☐ H) Subacute thyroiditis
- ☐ I) Thyroid cancer



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14. During a routine health maintenance examination, a 15-year-old girl says that she has used marijuana with friends on four occasions. She is healthy and doing well in school. She requests that her parents not be told about the marijuana use. In addition to counseling her about the adverse effects of marijuana use, which of the following is the most appropriate intervention at this time?

- ☐ A) Referral to a psychologist for treatment of substance abuse
- ☐ B) Scheduling a family conference
- ☐ C) Scheduling a follow-up examination with the patient
- ☐ D) Screening for human papillomavirus infection
- ☐ E) Urine toxicology screening



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15. A 56-year-old woman comes to the physician because of a 3-day history of an itchy, burning rash over the left side of her face. She has no history of serious illness and takes no medications. She is a retired schoolteacher. Her temperature is 37.1°C (98.8°F), pulse is 81/min, respirations are 14/min, and blood pressure is 138/80 mm Hg. A photograph of the rash is shown. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Topical triamcinolone
- ☐ B) Oral cephalexin
- ☐ C) Oral prednisone
- ☐ D) Oral trimethoprim-sulfamethoxazole
- ☐ E) Oral valacyclovir



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16. A 62-year-old woman comes to the physician because of a 6-month history of shortness of breath with exertion and leg swelling. Her last visit to a physician was 15 years ago. She takes no medications. Bilateral basilar crackles are heard on auscultation. Cardiac examination shows a dyskinetic, laterally displaced point of maximal impulse and an S_3 . There is 2+ pitting edema of the lower extremities. Laboratory studies show:

Serum	
Na ⁺	132 mEq/L
Cl ⁻	104 mEq/L
K ⁺	3.8 mEq/L
HCO ₃ ⁻	22 mEq/L
Urea nitrogen	14 mg/dL
Glucose	98 mg/dL
Creatinine	1.2 mg/dL
Urine	
Specific gravity	1.020
Protein	negative

Which of the following is the most likely explanation for this patient's hyponatremia?

- ☐ A) Baroreceptor-mediated ADH (vasopressin) secretion
- ☐ B) Ectopic production of ADH
- ☐ C) Inappropriate suppression of ADH
- ☐ D) Inappropriate suppression of ADH receptors
- ☐ E) Renal unresponsiveness to ADH



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17. A 12-year-old boy is brought to the physician for evaluation of bed wetting; he was toilet trained for daytime voiding at the age of 3 years but has continued to wet the bed three to five times weekly. He has not had constipation or stool incontinence. He also has difficulty waking up in the morning to get ready for school. His parents describe him as a "deep sleeper." His development is appropriate for age. He is at the 50th percentile for height, weight, and BMI. Vital signs are within normal limits. Physical examination shows no abnormalities. Which of the following is the most appropriate management for this patient to achieve nighttime continence?

- ☐ A) Water deprivation after 5 PM
- ☐ B) Enuresis alarm
- ☐ C) Desmopressin therapy
- ☐ D) Fluoxetine therapy
- ☐ E) Imipramine therapy



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18. A 77-year-old man with coronary artery disease comes to the physician for a follow-up examination. He underwent coronary artery bypass grafting 4 years ago. He has a 3-year history of intermittent claudication. His blood pressure measurements from his last eight visits show:

Number of Months Ago	Blood Pressure (mm Hg)
24	130/80
21	130/60
15	140/70
12	130/80
8	160/90
6	170/96
4	180/110
2	180/100

Current medications include aspirin and atorvastatin. His temperature is 36.5°C (97.7°F), pulse is 80/min, and blood pressure is 180/110 mm Hg. Heart sounds are normal. Dorsalis pedis and posterior tibial pulses are not palpable bilaterally. Serum studies show:

K ⁺	3.1 mEq/L
HCO ₃ ⁻	28 mEq/L
Urea nitrogen	20 mg/dL
Glucose	90 mg/dL
Creatinine	1.1 mg/dL

Which of the following is the most likely cause of this patient's increased blood pressure?

- ☐ A) Dissecting abdominal aortic aneurysm
- ☐ B) Essential hypertension
- ☐ C) Renal artery stenosis
- ☐ D) Renal failure
- ☐ E) White-coat hypertension



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19. An 11-year-old boy is brought to the physician because of a 2-day history of nasal congestion and mild sore throat; he has coughing and sneezing productive of green mucus. He has not had fever or ear pain. He has had no sick contacts. His temperature is 37°C (98.6°F), and pulse is 84/min and regular. Examination shows clear and mobile tympanic membranes. The tonsils are mildly enlarged and erythematous; there is no exudate. There is postnasal drainage in the posterior pharynx and copious yellow-green nasal drainage. The lungs are clear to auscultation. Heart sounds are normal. Which of the following is the most appropriate next step in management?

- ☐ A) Provide symptomatic care only
- ☐ B) Recommend sinus lavage in the clinic
- ☐ C) Culture of the nasal discharge
- ☐ D) X-ray of the sinuses
- ☐ E) Antiviral therapy
- ☐ F) Broad-spectrum antibiotic therapy



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20. A 15-year-old girl is brought to the physician because of left ear pain for 2 days. She competes in long-distance open-water swimming events and trains in the ocean several hours daily. Vital signs are within normal limits. Examination of the left ear shows erythema and edema of the auditory canal and a small amount of white discharge. Manipulation of the left pinna produces pain; the left tympanic membrane appears normal. After treating this patient for her acute condition, recommendation of which of the following is the most appropriate next step to prevent recurrence of this condition?

- ☐ A) Training in chlorine-treated pools only
- ☐ B) Daily use of topical benzocaine in the ear canal
- ☐ C) Daily use of alcohol-acetic acid ear drops
- ☐ D) Daily use of carbamide peroxide ear drops
- ☐ E) Daily use of hydrocortisone ear drops



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21. A 62-year-old man comes to the physician because of a 2-month history of shortness of breath with exertion. He also has had paroxysmal nocturnal dyspnea and orthopnea. He has type 2 diabetes mellitus and coronary artery disease; he had a myocardial infarction 3 years ago. Current medications include simvastatin and aspirin. His pulse is 100/min, respirations are 12/min, and blood pressure is 100/60 mm Hg. Crackles are heard on auscultation. There is an S_3 . An x-ray of the chest shows cardiomegaly, cephalization of blood flow, and Kerley B lines. Which of the following is the most appropriate pharmacotherapy to improve this patient's chances of survival?
- ☐ A) Angiotensin-converting enzyme (ACE) inhibitor
 - ☐ B) Calcium-channel blocking agent
 - ☐ C) Digitalis
 - ☐ D) Diuretic
 - ☐ E) Nitrates



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22. A 42-year-old woman with type 2 diabetes mellitus comes to the physician because of a lesion on her perineum for 3 months. Examination shows an ulcerated plaque on the perineal body. A biopsy specimen of the lesion shows invasive squamous cell carcinoma; there is extensive intraepithelial disease on the perineum. She asks how she could have acquired this cancer. Which of the following is the most likely underlying cause?

- ☐ A) Exposure to herpesvirus
- ☐ B) Exposure to human papillomavirus
- ☐ C) Exposure to lymphogranuloma venereum
- ☐ D) Prolonged periods of hyperglycemia that alter immune function
- ☐ E) Use of excessive amounts of talcum powder



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23. A healthy 37-year-old woman comes for a routine physical examination. She has a history of rheumatic fever as a child. Her vital signs are within normal limits. On cardiac examination, a left parasternal lift is palpated; a diastolic murmur at the apex and lower left sternal border and an S_4 are heard on auscultation. Which of the following is the most likely cause of this patient's findings?

- ☐ A) Aortic valve insufficiency
- ☐ B) Aortic valve stenosis
- ☐ C) Atrial septal defect
- ☐ D) Hypertrophic obstructive cardiomyopathy
- ☐ E) Mitral valve insufficiency
- ☐ F) Mitral valve prolapse
- ☐ G) Mitral valve stenosis
- ☐ H) Pulmonic valve insufficiency
- ☐ I) Pulmonic valve stenosis
- ☐ J) Tricuspid valve insufficiency
- ☐ K) Tricuspid valve stenosis
- ☐ L) Ventricular septal defect



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24. A 14-year-old boy with reactive airway disease is brought to the physician because of repeated episodes of wheezing over the past year. He has been hospitalized six times during this period and has missed numerous days of school. His wheezing is exacerbated by playing basketball or soccer. Medications include nebulized albuterol. He also has atopy; previous skin testing has shown allergies to pollens, milk protein, and ragweed. His temperature is 36.4°C (97.5°F), pulse is 70/min, respirations are 18/min, and blood pressure is 110/70 mm Hg. A few scattered wheezes are heard bilaterally. The remainder of the examination shows no abnormalities. His FEV₁ is 64% of predicted before bronchodilator therapy and 80% of predicted after bronchodilator therapy. Which of the following is the most appropriate recommendation to decrease this patient's school absenteeism because of reactive airway disease?

- ☐ A) Inhaled corticosteroid therapy
- ☐ B) Inhaled long-acting β_2 -adrenergic agonist therapy
- ☐ C) Oral antihistamine therapy
- ☐ D) Allergen desensitization
- ☐ E) Avoidance of soccer and basketball



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25. A previously healthy 14-year-old boy is brought to the physician because of a 1-week history of a rash over his neck, back, and chest. It is not itchy or painful. He has had no other symptoms. A photograph of the lesions is shown. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Topical benzoyl peroxide
- ☐ B) Topical clindamycin
- ☐ C) Topical hydrocortisone
- ☐ D) Topical mupirocin
- ☐ E) Topical permethrin
- ☐ F) Topical selenium sulfide
- ☐ G) Oral griseofulvin



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26. A 67-year-old man comes to the physician requesting a prescription for nicotine patches. He has smoked two packs of cigarettes daily for 50 years. He tried to quit abruptly several times without success. Five years ago, he used nicotine gum successfully for 3 months but then began smoking again. He has a 3-year history of stable angina pectoris. Examination shows no abnormalities. Which of the following is the most effective management for this patient?
- ☐ A) Advise the patient that he should not use transdermal nicotine because its use is contraindicated in patients with angina pectoris
 - ☐ B) Prescribe nicotine gum since it has been proven to be superior to transdermal nicotine in randomized trials in patients with heart disease
 - ☐ C) Prescribe transdermal nicotine at lower than recommended dosages and allow the patient to smoke two cigarettes daily to prevent relapse
 - ☐ D) Prescribe transdermal nicotine at lower than recommended dosages to prevent angina pectoris
 - ☐ E) Prescribe transdermal nicotine and recommend a behavior modification program



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27. A previously healthy 47-year-old man comes to the physician because of a 2-month history of severe headaches. Every night, he wakes up at 3 AM because of severe pain behind his left eye that feels as if he were being "stabbed by an ice pick." He then gets up and paces. During these episodes, he notices that his left pupil is smaller than the right and that his left eyelid droops. He has not had nausea. He has smoked two packs of cigarettes daily for 25 years. He usually drinks six beers nightly. His temperature is 37°C (98.6°F), pulse is 110/min, respirations are 20/min, and blood pressure is 145/90 mm Hg. Examination shows deep facial wrinkles. Fundusoscopic examination shows no papilledema. Which of the following is the most likely cause of these findings?

- ☐ A) Cluster headache
- ☐ B) Idiopathic intracranial hypertension
- ☐ C) Migraine without aura
- ☐ D) Nocturnal seizures
- ☐ E) Trigeminal neuralgia



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28. A 45-year-old man with gout has the sudden onset of extreme pain, swelling, and exquisite tenderness of the right knee; he cannot bear the pressure of wearing slacks. He has not recently taken any medication and previously had intractable diarrhea while taking colchicine. Which of the following is the most appropriate drug for treating this acute episode?

- ☐ A) Allopurinol
- ☐ B) Codeine
- ☐ C) Indomethacin
- ☐ D) Prednisone
- ☐ E) Probenecid



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29. A 42-year-old man comes to the physician for a routine life insurance examination. He feels well. He has smoked one pack of cigarettes daily for 25 years and drinks two to four alcoholic beverages daily. His work as a journalist sometimes takes him to developing countries, but he has not traveled abroad during the past year. He has not had any immunizations during adulthood except for tetanus toxoid 12 years ago. He has never had a blood transfusion. Examination shows no abnormalities. Laboratory studies show:

Hemoglobin	16.2 g/dL
Serum	
Albumin	3.8 g/dL
Total bilirubin	1.1 mg/dL
Alkaline phosphatase	95 U/L
AST	68 U/L
ALT	44 U/L
Ferritin	50 ng/mL

Which of the following is most likely to have prevented these laboratory abnormalities?

- ☐ A) Abstinence from alcohol
- ☐ B) Administration of diphtheria-tetanus toxoid
- ☐ C) Administration of hepatitis A vaccine
- ☐ D) Low-protein, branched-chain amino acid-enriched diet
- ☐ E) Smoking cessation
- ☐ F) Weekly phlebotomy



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30. An 82-year-old woman comes to the physician because her memory is "not as good as it used to be." She sometimes forgets where she left her keys, and she takes longer to recall words and names. She says her forgetfulness does not affect her daily functioning or diminish her enjoyment of her art and computer classes at the local senior center. She says she is not depressed. She has osteoporosis treated with alendronate and a calcium supplement containing vitamin D. Vital signs are within normal limits. Neurologic examination shows no focal findings. Her Mini-Mental State Examination score is 29/30. When asked to recall polysyllabic words, she uses multiple shorter words to describe the same concept. Her speech is normal in rate and rhythm. The remainder of the examination shows no abnormalities. Results of laboratory studies, including thyroid function tests and measurement of serum vitamin B₁₂ (cobalamin) concentration, are within the reference ranges. Which of the following is the most appropriate next step in management?

- ☐ A) Reassurance
- ☐ B) Speech therapy
- ☐ C) Neuropsychological testing
- ☐ D) MRI of the brain
- ☐ E) Methylphenidate therapy



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31. A 37-year-old man comes to the physician because of a 6-week history of epigastric pain. The pain is most severe 1 hour after eating and is relieved by over-the-counter antacids. He has had no fever, nausea, or weight loss. He has hypertension treated with hydrochlorothiazide. He does not smoke and drinks one glass of wine with dinner daily. His pulse is 76/min, and blood pressure is 136/88 mm Hg. There is mild epigastric tenderness without rebound or guarding. The stool is brown, and test for occult blood is negative. His hemoglobin concentration is 13.8 g/dL. Liver function tests show no abnormalities. Which of the following is the most appropriate next step in diagnosis?
- ☐ A) Esophageal pH monitoring
 - ☐ B) Measurement of serum amylase activity
 - ☐ C) Serum *Helicobacter pylori* antibody assay
 - ☐ D) Abdominal ultrasonography
 - ☐ E) Abdominal CT scan



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32. A 52-year-old man comes for a follow-up examination 2 weeks after sustaining an inferior wall myocardial infarction. Echocardiography before discharge showed an ejection fraction of 33%. Recovery has been uneventful except for decreased exercise tolerance. His temperature today is 37.2°C (99°F), pulse is 80/min, and respirations are 16/min. Examination shows no abnormalities. Which of the following is the most appropriate pharmacotherapy to prolong this patient's life?

- ☐ A) Digoxin
- ☐ B) Furosemide
- ☐ C) Isosorbide dinitrate
- ☐ D) Lisinopril
- ☐ E) Verapamil



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33. A 37-year-old woman, gravida 3, para 3, comes to the physician for a routine health maintenance examination. One year ago, her serum cholesterol concentration was 260 mg/dL, and she was started on a low-fat diet. She has no history of serious illness. Her Pap smears during the past 10 years have shown no abnormalities. Her last Pap smear was 1 year ago. Her mother has type 2 diabetes mellitus, and her father died of colon cancer at the age of 62 years. The patient does not smoke cigarettes or drink alcohol. She has been married for 15 years. She is 157 cm (5 ft 2 in) tall and weighs 77 kg (170 lb); BMI is 31 kg/m². Her pulse is 64/min, and blood pressure is 120/60 mm Hg. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in management?

- ☐ A) Colonoscopy
- ☐ B) 5-Hour glucose tolerance test
- ☐ C) Mammography
- ☐ D) Measurement of serum cholesterol concentration
- ☐ E) Pap smear



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34. A 77-year-old woman comes to the physician because of a 1-week history of decreased appetite, nausea, and occasional vomiting. She also reports fullness and discomfort in her abdomen and abdominal cramps that have awakened her on several occasions. Her last bowel movement was 5 days ago and was characterized by loose stool. Thirty years ago, she underwent hysterectomy for leiomyomata uteri. She has a 10-year history of intermittent urinary incontinence. Her only medication is oxybutynin. Colonoscopy 3 years ago showed no abnormalities. She appears frail. She is 157 cm (5 ft 2 in) tall and weighs 57 kg (125 lb); BMI is 23 kg/m². Her temperature is 37°C (98.6°F), pulse is 80/min and regular, respirations are 14/min, and blood pressure is 130/85 mm Hg. Examination shows warm, dry skin. The lungs are clear to auscultation and percussion. Cardiac examination shows no murmurs. The abdomen is soft and diffusely tender; there is no rebound or organomegaly. Rectal examination shows firm stool in the rectal vault. Which of the following is the most likely cause of this patient's current symptoms?

- ☐ A) Adhesions
- ☐ B) Colon cancer
- ☐ C) Diverticulitis
- ☐ D) Fecal impaction
- ☐ E) Ischemic bowel
- ☐ F) Volvulus



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35. An 82-year-old man comes to the physician because of an 8-month history of progressive urinary frequency and decreased urinary stream. He underwent transurethral resection of the prostate 11 years ago because of benign hyperplasia. He takes no medications. Examination shows a moderately enlarged prostate. His serum creatinine concentration has increased from 1.3 mg/dL 10 months ago to 2.4 mg/dL today. Which of the following is the most appropriate initial step in management?

- ☐ A) Measurement of postvoid residual volume
- ☐ B) Urine cytology
- ☐ C) Intravenous pyelography
- ☐ D) Retrograde urethrography
- ☐ E) CT scan of the abdomen



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36. A 47-year-old woman comes to the physician because of right wrist pain that began 2 days ago when she fell on her outstretched arm while playing tennis. She has had limited use of her hand and wrist since the injury occurred. She is otherwise healthy. There is tenderness to palpation over the radial and dorsal aspect of the right anatomic snuffbox. Muscle strength of the right fingers is 5/5, and sensation throughout the hand is intact. Plain x-rays of the hand and wrist show mild degenerative changes at the first carpometacarpal joint. Which of the following is the most appropriate next step in management?

- ☐ A) Reassurance that the pain should subside within 1 week
- ☐ B) Arthrography
- ☐ C) Cast immobilization
- ☐ D) Arthroscopy
- ☐ E) Surgical exploration and repair



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37. A 62-year-old woman comes to the physician because she is concerned about her risk for type 2 diabetes mellitus. Her mother and two sisters received the diagnosis of type 2 diabetes mellitus in their 50s. The patient is 157 cm (5 ft 2 in) tall and weighs 75 kg (165 lb); BMI is 30 kg/m². Her pulse is 80/min, and blood pressure is 130/70 mm Hg. Examination shows no other abnormalities. Laboratory studies show:

Hemoglobin A _{1c}	8.4%
Serum	
Cl ⁻	100 mEq/L
HCO ₃ ⁻	29 mEq/L
Urea nitrogen	30 mg/dL
Glucose	200 mg/dL
Creatinine	1.8 mg/dL

Which of the following medications is contraindicated in this patient?

- ☐ A) Acarbose
- ☐ B) Aspirin
- ☐ C) Glipizide
- ☐ D) Insulin
- ☐ E) Metformin



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38. An 82-year-old woman with dementia, Alzheimer type, is brought to the physician by her daughter for recommendations about driving restrictions. Since the patient moved in with her daughter 3 weeks ago, they have had several arguments because her daughter has not allowed the patient to drive. The patient last drove a car 2 years ago. She has no history of motor vehicle collisions. The patient's husband provided her daily care until he died of a myocardial infarction 3 weeks ago. The patient's only medication is galantamine, which she has taken for 4 years. Physical examination shows no abnormalities. Mental status examination shows a sad mood and reactive affect. She reports no suicidal thoughts. She is able to correctly repeat six digits forward, and she recalls one of three objects after 5 minutes; she is unable to correctly copy a rectangle or two intersecting pentagons. Which of the following is the most appropriate recommendation regarding this patient's driving?
- ☐ A) The patient should be allowed to drive only with supervision
 - ☐ B) The patient should be allowed to drive short distances only
 - ☐ C) The patient should be allowed to drive until her driving skills are formally assessed
 - ☐ D) The patient should not have access to car keys until her driving skills are formally assessed
 - ☐ E) The patient's driver license should be revoked



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39. A 47-year-old man comes to the physician because of a 2-day history of progressive sore throat and a 1-day history of a nonproductive cough. He can eat without difficulty. He has not had fever. He has no history of serious illness and takes no medications. He has no known drug allergies. He lives alone and works the night shift as a security guard. His temperature is 37.4°C (99.3°F), pulse is 80/min, and blood pressure is 140/90 mm Hg. Examination shows no lymphadenopathy. The pharynx is erythematous without exudates. Abdominal examination shows no abnormalities. Which of the following is the most appropriate next step in management?

- ☐ A) Warm saline gargle
- ☐ B) Monospot test
- ☐ C) Rapid streptococcal test
- ☐ D) Throat culture for streptococci
- ☐ E) Penicillin therapy



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40. A previously healthy 82-year-old woman comes to the physician because of generalized weakness and a sore tongue for 1 month. She lives by herself. Her diet has consisted mainly of tea and homemade bread since her husband died 6 months ago. Examination shows pallor, mild jaundice, and a smooth, red tongue. There is no lymphadenopathy or organomegaly. Neurologic examination shows no focal findings. Laboratory studies show:

Hematocrit	26%
Mean corpuscular volume	120 μm^3
Leukocyte count	2500/ mm^3
Reticulocyte count	0.2%
Platelet count	20,000/ mm^3

This patient most likely has which of the following nutritional deficiencies?

- ☐ A) Folic acid
- ☐ B) Niacin
- ☐ C) Vitamin B₁ (thiamine)
- ☐ D) Vitamin B₆
- ☐ E) Vitamin B₁₂ (cobalamin)



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The following vignette applies to the next **2** items. The items in the set must be answered in sequential order. Once you click **Proceed to Next Item**, you will not be able to add or change an answer.

A 55-year-old woman comes to the office because of a 3-month history of fatigue. She typically walks to work daily but has been driving to work during this time because of decreased energy. She has hypertension and gastroesophageal reflux disease well controlled with lisinopril and omeprazole. Her only other medication is naproxen daily during the past 3 months for arthritis of the knees. Five years ago, colonoscopy showed two hyperplastic polyps, which were removed. Menopause occurred 2 years ago; she has periodic hot flashes. Vital signs are within normal limits. Examination shows no abnormalities. Laboratory studies are shown:

Hemoglobin	10.4 g/dL
Hematocrit	32%
Mean corpuscular volume	80 μm^3
Leukocyte count	4200/mm ³ with a normal differential

Results of serum lipid studies, liver function tests, and erythrocyte sedimentation rate are within the reference ranges.

41. Item 1 of 2

Measurement of which of the following is the most appropriate next step in diagnosis?

- ☐ A) Erythrocyte protoporphyrin concentration
- ☐ B) Reticulocyte count
- ☐ C) Serum ferritin concentration
- ☐ D) Serum folate concentration
- ☐ E) Serum vitamin B₁₂ (cobalamin) concentration

Proceed to Next Item

You must select "Proceed to Next Item" to view Item 2 of 2



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Leukocyte count

4200/mm³ with a normal differential

Results of serum lipid studies, liver function tests, and erythrocyte sedimentation rate are within the reference ranges.

41. Item 1 of 2

Measurement of which of the following is the most appropriate **next** step in diagnosis?

- ☐ A) Erythrocyte protoporphyrin concentration
- ☒ B) Reticulocyte count
- ☐ C) Serum ferritin concentration
- ☐ D) Serum folate concentration
- ☐ E) Serum vitamin B₁₂ (cobalamin) concentration

42. Item 2 of 2

The patient's serum ferritin concentration is 5 ng/mL. She is advised to discontinue naproxen therapy and begin iron supplementation twice daily. One month later, she returns to the office for a follow-up examination. She has adhered to her medication regimen. Laboratory studies today are shown:

Hemoglobin	10.5 g/dL
Hematocrit	33%
Mean corpuscular volume	80 μm^3
Serum ferritin	6 ng/mL

Which of the following is the most appropriate next step in management?

- ☐ A) Bone marrow biopsy
- ☐ B) Colonoscopy
- ☐ C) CT scan of the abdomen
- ☐ D) Esophageal pH monitoring
- ☐ E) Observation and follow-up examination in 1 month

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43. The wife of a 72-year-old patient calls the physician because she is concerned that it is no longer safe for her husband to drive; he has had two motor vehicle collisions within the past month. She asks the physician to tell her husband to stop driving at his scheduled appointment the next day. Which of the following is the most appropriate action for the physician?
- ☐ A) Ask the wife to accompany her husband to his appointment and express her concerns in her husband's presence
 - ☐ B) Ask the wife to contact the Department of Motor Vehicles about her concerns
 - ☐ C) Tell the patient that he must take a driving test
 - ☐ D) Contact the Department of Motor Vehicles and express the wife's concerns
 - ☐ E) Schedule a CT scan of the head prior to the next visit



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44. A 25-year-old woman comes to the physician because of a 2-week history of neck pain, palpitations, and sweating; she has had a 2.3-kg (5-lb) weight loss during this period. Her pulse is 120/min, and blood pressure is 150/70 mm Hg. Examination shows an enlarged tender thyroid gland. A grade 2/6, systolic ejection murmur is heard at the left sternal border. Her serum thyroxine (T_4) concentration is 15 $\mu\text{g/dL}$. A thyroid scan shows decreased uptake. Which of the following is the most likely diagnosis?

- ☐ A) Exogenous thyroid ingestion
- ☐ B) Graves disease
- ☐ C) Subacute thyroiditis
- ☐ D) Thyroid cancer
- ☐ E) Toxic multinodular goiter



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45. A 27-year-old man comes to the physician for a routine health maintenance examination in April. He feels well and has no history of serious illness. He received all childhood immunizations. He has smoked one pack of cigarettes daily for 5 years. Examination shows no abnormalities. Which of the following vaccines are most appropriate to administer at this visit?

	23-Valent Pneumococcal	<i>Haemophilus influenzae</i>	Influenza Virus
☐ A)	No	no	no
☐ B)	No	yes	yes
☐ C)	Yes	no	yes
☐ D)	Yes	yes	no
☐ E)	Yes	yes	yes



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46. A previously healthy 47-year-old woman comes to the physician because of a 1-week history of low back pain that radiates to the left buttock and along the lateral side of the left leg. Her symptoms began 1 day after moving furniture in her home. She has not had fever, chills, constipation, or urinary incontinence. Examination shows diffuse tenderness over the paraspinal muscles of the lumbosacral area and no tenderness over the spine. Range of motion of the left lower extremity is limited by pain. Deep tendon reflexes are normal, muscle strength is intact, and sensory examination shows no abnormalities. Which of the following is the most appropriate next step in management?
- ☐ A) Bed rest for 1 week
 - ☐ B) CT scan of the lumbosacral spine
 - ☐ C) Epidural corticosteroid injection
 - ☐ D) Oral ibuprofen therapy
 - ☐ E) Oral prednisone therapy



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47. A 47-year-old woman comes to the physician for a routine health maintenance examination. She has 11-year-old twin boys. During a conversation about her family, she becomes tearful and states, "I don't know what is wrong with me. I am so irritable with my boys, and they are good kids. I am such a bad mother." She says that she has difficulty concentrating during the day and falling asleep at night. She no longer is interested in activities that she used to enjoy. She has no history of serious illness and takes no medications. She drinks one glass of wine several times weekly to help her relax before she goes to bed. Vital signs are within normal limits. Physical examination shows no abnormalities. Which of the following is the most appropriate physician response to this mother's concern about parenting?

- ☐ A) "Don't be so hard on yourself. All 11-year-old boys make their parents irritable, and you have two boys to deal with."
- ☐ B) "I would recommend a sleeping pill. If you sleep more, you'll be less irritable."
- ☐ C) "Sometimes when people are as upset as you are, they think about suicide. Have you thought about ending your life?"
- ☐ D) "Would you like to try an antianxiety medication? It may help you cope."
- ☐ E) "You might want to consider abstaining from alcohol, since it may be making you more irritable."



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48. An asymptomatic 42-year-old man with a 1-month history of essential hypertension comes for a follow-up examination. He has been moderately obese for 20 years and follows a high-calorie, high-fat diet. He drinks four to five cups of coffee daily. He has smoked one pack of cigarettes daily for 20 years. He drinks one beer at dinner almost daily. He has a sedentary lifestyle. He is 175 cm (5 ft 9 in) tall and weighs 100 kg (220 lb); BMI is 33 kg/m². His pulse is 78/min, and blood pressure is 148/94 mm Hg. Examination shows no other abnormalities. Which of the following is the most effective method to decrease his blood pressure?

- ☐ A) Abstinence from alcohol
- ☐ B) Abstinence from coffee
- ☐ C) Smoking cessation
- ☐ D) Weight loss of 9 kg (20 lb)
- ☐ E) Biofeedback
- ☐ F) Vitamin E supplementation



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49. A 57-year-old man comes to the physician because of a 3-month history of intermittent chest pain with eating. Occasionally, he has the sensation that food is stuck in his middle to lower chest; the sensation is relieved after he drinks a large amount of water. He has a 20-year history of gastroesophageal reflux disease treated with a proton pump inhibitor. He drinks three cups of coffee daily and drinks one martini with dinner every night. He often works late and arrives home 1 hour before bedtime. He is 188 cm (6 ft 2 in) tall and weighs 116 kg (255 lb); BMI is 33 kg/m². Examination shows no abnormalities. Which of the following is the most appropriate next step in diagnosis?
- ☐ A) Barium swallow
 - ☐ B) CT scan of the chest
 - ☐ C) Esophageal manometry
 - ☐ D) Upper endoscopy
 - ☐ E) 24-Hour monitoring of esophageal pH



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50. One hundred patients participate in a health education program designed to teach health benefits of a high-fiber, low-cholesterol diet. The effectiveness of this program is most appropriately measured by which of the following?

- ☐ A) Assessment of health status of each participant
- ☐ B) Attendance rates of participants
- ☐ C) Change in mortality rate from ischemic heart disease among participants
- ☐ D) Change in knowledge and dietary habits of each participant
- ☐ E) Change in morbidity among participants



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